ENGINEERING RESEARCH NOTES

Task	Status
Research all about BLE antenna	✓ Complete
Research crystal / no-crystal decision — document trade-off for 40 MHz XTAL	✓ Complete

Task 1 — Research: BLE Antenna for ESP32-S3

Bluetooth Low Energy (BLE) and WiFi both operate in the 2.4 GHz ISM band (2.400–2.4835 GHz). A single antenna on the ESP32-S3 serves both protocols simultaneously. This section documents all key findings from the antenna research, covering antenna types, matching network theory, layout rules, and validation methods.

1.1 Why Antenna Design Matters for BLE

The ESP32-C3's RF output impedance is approximately 50 Ω. This means you cannot connect any antenna directly to the LNA_IN/RF pin without an impedance-matching network between the chip and the antenna. An unmatched antenna causes signal reflection (high S11), which wastes transmit power, reduces receive sensitivity, and can in severe cases damage the RF front-end of the chip. A well-designed BLE antenna with proper matching achieves:

- Return loss (S11) < −10 dB across the full 2400–2483.5 MHz band
- Resonant frequency centred at approximately 2441 MHz (mid-band)
- Bandwidth ≥ 83.5 MHz to cover the entire BLE and WiFi 2.4 GHz band
- Radiation efficiency sufficient for the target communication range

1.2 Antenna Type Comparison

Antenna Type	Size	Performance	BOM Cost	Best For
PCB Trace IFA	~4×20 mm	High gain 3.5–4 dBi	Free (PCB trace)	Max range; boards ≥ 30×30 mm ground plane
PCB Trace MIFA	~7×11 mm	Good 2–3 dBi	Free (PCB trace)	BLE IoT sensors; space-constrained boards
Chip Antenna ✓	2–5 mm	Good 1–3 dBi	$0.30–$1.50	Small boards; limited ground plane; certified designs
External (U.FL/SMA)	Off-board	Excellent >5 dBi	$1–$5 + cable	Maximum range; products where PCB space is critical
MIFA Module	Module incl.	Good; pre-certified	Module cost	Fastest time-to-market; no RF certification needed

1.3 Johanson 2450AT18A100E — Specific Chip Antenna

Key Specifications

Parameter	Value	Design Note
Frequency	2.4–2.5 GHz	Covers full BLE and WiFi 2.4G band
Impedance	50 Ω	Requires matching network from 35+j10 Ω ESP32-S3 RF output
Package	0805 chip (2.0×1.25 mm)	SMD; reflow-solderable; low BOM cost
Bandwidth	≥ 200 MHz at S11 < −10 dB	Comfortably covers 2.4–2.5 GHz with margin
Ground plane sensitivity	Moderate — chip antenna preferred for small boards	Less sensitive than PCB trace IFA; works on boards < 30×30 mm
Keepout requirement	3 mm on 3 sides; 5 mm in front of radiating element	No copper on ANY layer in keepout zone — critical

Reference Matching Network (from Johanson Datasheet)

The Johanson 2450AT18A100E reference design uses a simple two-element L-network between the ESP32-C3 RF pin and the antenna feed:

LNA_IN/RF pin → L1 (2.7 nH series inductor) → Antenna feed point

C2 (1.0 pF shunt capacitor to GND) — placed at antenna feed side

C1 (DNP — Do Not Populate) — shunt capacitor before L1 for tuning

It provides a tuning option during RF bring-up using a VNA without requiring a PCB respin. If the S11 measurement shows impedance drift, you populate C1 with the appropriate value to shift the resonance back to 2441 MHz.

1.4 PCB Layout Rules — Critical Requirements

A. Antenna Keepout Zone

This is the single most common reason chip antennas fail in production designs. The keepout zone around the Johanson 2450AT18A100E must be enforced on all PCB copper layers, not just the top layer:

Rule	Requirement	Consequence if Violated
Ground pour under antenna	NONE — clear all copper on all layers	Antenna detuned by 100–200 MHz; S11 > −3 dB; range drops to centimetres
Clearance in front of antenna	≥ 5 mm no copper	Pattern distortion; VSWR increases

Clearance on three remaining sides	≥ 3 mm no copper	Partial detuning; sensitivity degradation
Ground plane minimum size	≥ 25×18 mm continuous pour	Reduced efficiency; lower gain; reduced range
Crystal / high-frequency components	Keep ≥ 8 mm away from antenna	Crystal harmonics couple into antenna; spurious emissions

B. RF Trace from LNA_IN to Matching Network

The trace connecting the ESP32-C3 LNA_IN/RF pin to the L1/C1/C2 matching network must be a controlled-impedance 50 Ω coplanar waveguide (CPW) trace. On standard 1.6 mm FR4 PCB with outer copper layer:

- Trace width: approximately 2.0–2.4 mm (calculate for your exact stackup)
- Gap to ground pour on each side: 0.2 mm
- Trace length: as short as possible — ideally < 5 mm
- No vias in the RF trace path
- Use rounded 45° bends, never 90° corners — sharp corners cause impedance discontinuities

✓ **DESIGN TIP:** Use the KiCad RF Toolkit plugin to generate 50 Ω coplanar waveguide traces automatically based on your board stackup parameters. This eliminates the manual calculation error risk.

C. Component Separation

Espressif's ESP32-C3 hardware design guidelines explicitly require the following separations on the PCB:

- Antenna must be placed away from the 40 MHz crystal oscillator (the crystal harmonics can couple into the RF path)
- No high-frequency signal traces (USB D+/D−, UART, SPI, SDIO) may route near the RF trace
- UART signal lines must be surrounded by ground copper and ground vias if routed near the RF section
- Antenna feed point should be at or near the board edge for maximum radiation efficiency

1.5 Validation — How to Verify Antenna Performance

Bench Validation Procedure

Step	Action	Tool Required	Pass Criterion
1	Calibrate VNA at the measurement port	NanoVNA or professional VNA	Calibration complete, SOL (Short-Open-Load)
2	Connect VNA to RF pin through the matching network	NanoVNA + SMA pigtail	Stable connection, no mechanical movement
3	Sweep S11 from 2.0 to 3.0 GHz	NanoVNA	S11 < −10 dB across 2.400–2.483 GHz

4	Identify resonant frequency (S11 minimum)	NanoVNA	Resonant dip centred near 2441 MHz
5	Compare RSSI against reference ESP32 module at known distance	Two ESP32-S3 boards + nRF Connect app	RSSI within 3–5 dB of reference module
6	Test BLE range in application environment	Phone + nRF Connect	Achieves required range with RSSI > −90 dBm

Task 2 — Research: Crystal vs No-Crystal Decision for 40 MHz XTAL

Clock source selection is a fundamental design decision in embedded systems, directly influencing timing accuracy, wireless performance, power consumption, cost, and reliability. The ESP32-C3 supports both an external 40 MHz crystal oscillator (XTAL) and an internal RC oscillator. This research evaluates the architectural differences, performance implications, and application suitability of these two approaches. The analysis shows that while the internal oscillator simplifies design and reduces cost, the external crystal provides superior frequency stability essential for RF communication and precise timing applications.

Modern wireless microcontrollers depend heavily on stable clock sources for:

- RF modulation/demodulation
- Protocol timing (Wi-Fi, BLE)
- CPU execution and peripheral synchronization

The ESP32-C3 allows two primary clocking strategies:

- External 40 MHz Crystal (XTAL)
- Internal RC Oscillator (No external crystal)

Each presents trade-offs across performance, cost, and complexity.

1. Clock Architecture Overview

1.1 External 40 MHz Crystal

A quartz crystal connected to the XTAL pins provides:

- High frequency stability
- Low phase noise
- Temperature compensation (inherent to quartz)

Typical configuration:

- 40 MHz crystal
- Load capacitors (≈18–22 pF)
- Short, symmetric PCB traces

1.2 Internal RC Oscillator

The internal oscillator is:

- Integrated on-chip
- No external components required
- Based on resistive-capacitive timing

Characteristics:

- Lower accuracy
- Temperature and voltage dependent
- Subject to calibration mechanisms

2. Performance comparison

Parameter	External 40 MHz Crystal	Internal RC Oscillator
Frequency Accuracy	±10–30 ppm	±1% to ±5%
Temperature Stability	High	Low
Startup Time	Moderate	Fast
Phase Noise	Low	High
RF Performance	Excellent	Limited
Power Consumption	Slightly higher	Lower
Cost	Higher (extra parts)	Lower
PCB Complexity	Moderate	Minimal

3. Impact on Wireless Connection

3.1 RF Sensitivity and Stability

Wireless protocols such as Wi-Fi and BLE require precise carrier frequency alignment.

- External crystal ensures tight frequency tolerance
- Internal RC oscillator introduces frequency drift

Implications:

- Reduced receiver sensitivity
- Increased packet loss

- Potential failure to meet protocol specifications

3.2 Regulatory Compliance

Wireless devices must meet standards (e.g., FCC, CE):

- Frequency deviation limits
- Spectral mask requirements

External crystals are typically mandatory for certification-grade designs.

4. Timing and Peripheral Accuracy

External Crystal

- Accurate timers and delays
- Stable UART baud rates
- Reliable ADC/DAC timing

Internal Oscillator

- Timing drift over temperature
- UART errors at high baud rates
- Reduced precision in time-critical systems

5. Power Consumption Considerations

Internal oscillator:

- Lower static power
- Faster wake-up from sleep

External crystal:

- Slightly higher consumption
- Better stability allows efficient RF operation

In many wireless cases, better RF efficiency offsets crystal power cost.

6. Environmental Sensitivity

Factor	External Crystal	Internal Oscillator
Temperature	Stable	Drifts significantly
Voltage variation	Minimal effect	Noticeable drift
Aging	Very slow	Moderate

7. Design Complexity

External Crystal Requires:

- Crystal component
- Load capacitors
- Careful PCB layout
- EMI considerations

Internal Oscillator Requires:

- No external components
- Minimal layout effort

8. Cost Analysis

Cost Element	External Crystal	Internal
BOM cost	Higher	Lower
Assembly	More complex	Simpler
PCB area	Increased	Reduced

9. Reliability Considerations

Crystals can fail due to:

- mechanical stress
- poor soldering

- vibration

Internal oscillators:

- no external failure points
- but less predictable behavior

10. Calibration and Compensation

Internal oscillators may use:

- software calibration
- temperature compensation

However:

They cannot fully match crystal stability, especially for RF systems.

11. Application Suitability

External 40 MHz Crystal (Recommended for)

- Wireless communication systems
- High-precision timing applications
- Regulatory-compliant products
- Industrial and commercial devices

Internal Oscillator (Suitable for)

- Low-cost designs
- Non-RF applications
- Prototyping and testing
- Ultra-low component count systems

12. Trade off Summary

Criterion	Best Choice
Accuracy	External Crystal
Cost	Internal Oscillator
Simplicity	Internal Oscillator
RF Performance	External Crystal
Reliability (system-level)	External Crystal
Power (idle)	Internal Oscillator